Operations Research: Essentials

Every theorem and method of Lectures 1–5, each with a practice question, its solution right underneath and the gotcha; plus the full method for reading everything off a final tableau or a basis B

Operations Research, midterm

2026-11-02. Every number was recomputed exactly (rational basis enumeration, `verify_essentials.py`).

How to use it. Read the purple *Theorem* boxes to know what you may quote and what hypothesis it needs. Read the blue *Method* box, cover the green *Solution*, do the orange *Practice*, compare, then read the red *Gotcha*. Part V is the one you asked for: a dictionary of everything a final tableau or a basis matrix tells you, in both directions.

Conventions of the course (and of this guide). Maximisation. Equational form $\max\{c^\top x : Ax = b,\ x \geq 0\}$ with $\operatorname{rank} A = m$. A basis B is an ordered list of m independent columns; $x_B = B^{-1}b$, $x_N = 0$. The tableau is written *equation style*: basic variables on the left, nonbasic on the right,

$$x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj} x_j, \qquad z = z_0 + \sum_{j \in N} r_j x_j.$$

A reduced cost $r_j > 0$ means "increasing x_j improves z". $y^\top = c_B^\top B^{-1}$ are the multipliers (shadow prices, dual solution).

The running example, used by most practices. A workshop makes two products, $x_1, x_2 \geq 0$ units per day. Resource 1: $2x_1 + x_2 \leq 10$ hours. Resource 2: $x_1 + x_2 \leq 6$ hours. The market takes at most 5 units of product 2: $x_2 \leq 5$. Profits 4 and 3 (hundreds of MAD).

$$\max\ z = 4x_1 + 3x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \leq 10, \quad x_1 + x_2 \leq 6, \quad x_2 \leq 5, \quad x \geq 0.$$

Slacks s_1, s_2, s_3; variable order $(x_1, x_2, s_1, s_2, s_3)$. Answer, for checking: $(x_1, x_2) = (4, 2)$, $z^\star = 22$, $y = (1, 2, 0)$.

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Part I

Station 1: Model

1 Formulate a linear program

Method

(a) **Variables**: one per decision, defined in words with unit and domain ("x_1 = units of product 1 made per day, $x_1 \geq 0$, continuous because divisible").

(b) **Objective**: max or min of a linear expression; state its unit.

(c) **Constraints**: one row per limit, read the coefficients *down the column* of each variable in the data table; state the unit of every row; add the domain constraints.

(d) **Class**: all linear and continuous ⇒ LP; integrality ⇒ IP; a product of variables ⇒ NLP. A percentage requirement ("at least 25 % of the units are product 2") is linear after clearing the denominator: $x_2 \geq 0.25(x_1 + x_2) \iff 0.25x_1 - 0.75x_2 \leq 0$.

Practice

Given. A workshop makes products 1 and 2. Product 1 needs 2 h on machine A and 1 h on machine B; product 2 needs 1 h on A and 1 h on B. Machine A has 10 h per day, machine B 6 h. At most 5 units of product 2 can be sold per day. Profits: 400 and 300 MAD per unit. Everything made is sold; fractional units are fine for planning.

Your task. Formulate, with units, and name the class.

Solution

Variables: $x_1, x_2 \geq 0$ units made per day (continuous). Objective in hundreds of MAD per day: $\max 4x_1 + 3x_2$. Rows: machine A hours $2x_1 + x_2 \leq 10$; machine B hours $x_1 + x_2 \leq 6$; market, units of product 2, $x_2 \leq 5$. LP: linear objective, linear rows, continuous variables. (This is the running example.)

Gotcha

"x_1 = product 1" is not a definition: say *how many units, per what period.* Changing the objective's unit (MAD to hundreds of MAD) changes z only, never the rows. "At most 5 can be sold" is a constraint on x_2, not a reason to add a sales variable, because every unit made is sold.

Part II

Station 2: Represent

2 Canonical and equational forms; substitutions

Theorem: Canonical and equational forms are equivalent

Canonical: $\max\{c^\top x : Ax \leq b,\ x \geq 0\}$. Equational: $\max\{c^\top x : Ax = b,\ x \geq 0\}$. A $\leq$ row gets a slack $s \geq 0$ added; a $\geq$ row gets a surplus subtracted; an $=$ row stays. The original decisions are unchanged and the optimal values coincide (a feasible point of one form gives one of the other

with the same value).

Method

(a) **Domains first.** $t \geq \ell$: set $u = t - \ell \geq 0$. $t \leq 0$: $t = -u$. t free: $t = u - v$, $u, v \geq 0$. Substitute everywhere, including the objective; **keep the constant** that appears.

(b) **Rows.** $a^\top x \leq \beta$: $a^\top x + s = \beta$. $a^\top x \geq \beta$: $a^\top x - s = \beta$. Minimisation: maximise $-f$ and negate the value at the end.

(c) **Order the variables** and write A, b, c in that order. Say the order.

Practice

Given. min $3x - 2y$ s.t. $x + y \geq 4$, $2x - y \leq 6$, $x \geq 1$, $y \in \mathbb{R}$.

Your task. Write an equivalent equational maximisation with nonnegative variables, give the variable order, A, b, c, and say how to recover the original optimal value.

Solution

$u = x - 1 \geq 0$, $y = v - w$ with $v, w \geq 0$. Objective: $\min 3(u+1) - 2(v-w) = \min 3u - 2v + 2w + 3$, so maximise $z = -3u + 2v - 2w - 3$. Rows: $(u+1) + (v-w) - s_1 = 4 \Rightarrow u + v - w - s_1 = 3$; $2(u+1) - (v-w) + s_2 = 6 \Rightarrow 2u - v + w + s_2 = 4$. Order (u, v, w, s_1, s_2):

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 & 0 \\ 2 & -1 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad c = (-3, 2, -2, 0, 0)^\top, \quad \text{constant } -3.$$

Original minimum $= -(z^\star)$ where $z^\star$ is the maximum of $c^\top x$ *plus* the constant: min $= -(z^\star_{\text{no const}} - 3)$. Say it in one line.

Gotcha

The surplus is *subtracted*. The constant from a substitution is dropped by half the class: the optimal *value* is off by it even though the optimal *point* is right. Multiplying a row by -1 to get $b \geq 0$ is allowed, but it flips the sign of that row's slack column in A: say that you did it.

3 Redundancy, consistency and rank

Theorem: Redundancy test; rank

An inequality is redundant if a **nonnegative combination** of the other rows (plus $x \geq 0$) implies it; removing it leaves the feasible set unchanged. For an equality system, elimination on $[A \mid b]$: a row $0 = 0$ can be dropped, a row $0 = \beta \neq 0$ means infeasible, the number of pivots is rank A. Slack columns form I_m, so after adding slacks rank $A = m$ automatically.

Method

To **prove redundancy**: exhibit the weights $\lambda_i \geq 0$ with $\sum \lambda_i a_i \leq a$ componentwise and $\sum \lambda_i b_i \leq \beta$ (then use $x \geq 0$ for the dominated coefficients). To **disprove** it: exhibit a point satisfying every other row but violating this one.

Practice

Given. The running example plus a fourth row $3x_1 + 2x_2 \leq 17$.

Your task. (a) Is the new row redundant? Prove it. (b) Is $x_2 \leq 5$ redundant? Prove your

answer.

Solution

(a) Yes: $1 \cdot (2x_1 + x_2 \le 10) + 1 \cdot (x_1 + x_2 \le 6)$ gives $3x_1 + 2x_2 \le 16 \le 17$. (b) No: $(0, 6)$ satisfies $2(0)+6 \le 10$, $0+6 \le 6$, $x \ge 0$ but violates $x_2 \le 5$. So the feasible sets differ, and the row cannot be dropped.

Gotcha

"It looks weaker" is not a proof. Weights must be ≥ 0 (a negative weight flips the inequality). Redundancy of an inequality never changes the optimal value, so "removing a redundant constraint changes $z^\star$" is false.

Part III

Station 2b: Bases and basic feasible solutions

4 Basis, BFS, enumeration, certificate

Theorem: Basis and BFS; fundamental theorem

With $\operatorname{rank} A = m$, a **basis** is an ordered choice of m linearly independent columns $B = [A_{j_1} \cdots A_{j_m}]$. Its basic solution: $x_N = 0$, $x_B = B^{-1}b$. It is a **BFS** iff $B^{-1}b \ge 0$; **nondegenerate** iff all m basic values are > 0. A BFS has *at most* m nonzero coordinates (basic ones may be 0).

Fundamental theorem. If the equational LP is feasible and the objective is bounded above, an optimal BFS exists. Hence enumerating the $\le \binom{n}{m}$ bases and comparing the feasible ones is a correct (slow) algorithm.

Used for. justifying enumeration; "a BFS has exactly m nonzeros" is **false** (degeneracy); several bases can give the same point.

Proof idea. From any optimal solution, remove a positive coordinate by moving along a direction in the kernel of the positive columns until a coordinate hits 0; the objective cannot decrease (otherwise unbounded in the other direction); induction on the support.

Method

(a) Rank: spot I_m in the slack columns, or eliminate.

(b) For each column choice: **independence** first (two columns proportional? $\det = 0$?). If dependent, it is not a basis at all.

(c) Set the other variables to 0, solve the $m \times m$ system by substitution, write the *full* vector.

(d) Feasible iff all coordinates ≥ 0. Compute z only for the feasible ones. Best feasible = optimum (fundamental theorem).

(e) **Certificate**: a nonnegative combination of the rows that bounds $c^\top x$ by $z^\star$ (that is the dual solution, Station 4).

Practice

Given. $\max 2x_1 + 3x_2$ s.t. $x_1 + 2x_2 + s_1 = 7$, $3x_1 + x_2 + s_2 = 6$, all variables ≥ 0. Order (x_1, x_2, s_1, s_2).

Your task. (a) rank A. (b) Table of the six column pairs: basis or not, full basic solution, feasible, z. (c) Optimal value, proved by a combination of the rows. (d) Unique?

Solution

(a) Columns 3, 4 are I_2, so $\operatorname{rank} A = 2$. (b) No pair is proportional, so all six are bases.

basic	x_1	x_2	s_1	s_2	feasible, z
x_1, x_2	1	3	0	0	yes, 11
x_1, s_1	2	0	5	0	yes, 4
x_1, s_2	7	0	0	-15	no
x_2, s_1	0	6	-5	0	no
x_2, s_2	0	$\frac{7}{2}$	0	$\frac{5}{2}$	yes, $\frac{21}{2}$
s_1, s_2	0	0	7	6	yes, 0

(x_1, x_2: $x_1 + 2x_2 = 7$, $3x_1 + x_2 = 6 \Rightarrow 5x_1 = 5$.) (c) $z^\star = 11$ at $(1, 3, 0, 0)$. Certificate: $\frac{7}{5}\,\text{row}_1 + \frac{1}{5}\,\text{row}_2$: $(\frac{7}{5}+\frac{3}{5})x_1 + (\frac{14}{5}+\frac{1}{5})x_2 + \frac{7}{5}s_1 + \frac{1}{5}s_2 = \frac{49}{5}+\frac{6}{5} = 11$, i.e. $2x_1 + 3x_2 = 11 - \frac{7}{5}s_1 - \frac{1}{5}s_2 \le 11$ for every feasible point. (d) Unique: $z = 11$ forces $s_1 = s_2 = 0$ (strictly negative coefficients), and then the rows give $(1, 3)$.

Gotcha

A dependent pair is "not a basis", not "infeasible". "The best of the feasible ones" proves optimality only through the fundamental theorem; if the question says *prove*, give the row combination. Where do the weights $\frac{7}{5}, \frac{1}{5}$ come from? They are $y^\top = c_B^\top B^{-1}$ of the optimal basis (Part V).

Part IV

Station 3: Simplex

5 The theorems behind one simplex step

Theorem: Tableau of a basis

For a basis B, define $r^\top = c^\top - c_B^\top B^{-1} A$. For *every* x with $Ax = b$ (feasible or not):

$$x_B = B^{-1}b - B^{-1}A_N x_N, \qquad c^\top x = c_B^\top B^{-1} b + \sum_{j \in N} r_j x_j, \qquad r_B = 0.$$

The tableau is unique: it depends on B only, not on the pivots that reached it.

Used for. everything in Part V; the objective row as a proof.

Proof idea. $Ax = Bx_B + A_N x_N$, solve for x_B, substitute into $c^\top x$.

Theorem: Basic direction and ratio test

Raise nonbasic x_j by $\theta \ge 0$, other nonbasics fixed at 0: direction $d_j = 1$, $d_B = -B^{-1}A_j$ (the displayed column of x_j), $Ad = 0$, gain $c^\top d = r_j$ per unit. Feasibility $x_B + \theta d_B \ge 0$ allows

$$\theta_{\max} = \min_{k:\ d_{B(k)} < 0} \frac{x_{B(k)}}{-d_{B(k)}}.$$

If no $d_{B(k)} < 0$ and $r_j > 0$: **unbounded**. If $\theta_{\max} = 0$: **degenerate** step (basis changes, point does not). Otherwise the pivot exchanges x_j with a row attaining the minimum and the new point is a BFS with z increased by $\theta_{\max} r_j$.

Theorem: Optimality, uniqueness, termination

(1) If the basic solution is feasible and all nonbasic $r_j \le 0$, it is optimal (every feasible x has $c^\top x = z_0 + \sum r_j x_j \le z_0$).

(2) Converse only under nondegeneracy: an optimal *nondegenerate* BFS has all $r_j \le 0$. At a degenerate optimum a basis may show $r_j > 0$ with $\theta_{\max} = 0$.

(3) If all nonbasic $r_j < 0$ strictly, the optimum is unique. If the BFS is nondegenerate and some nonbasic $r_j = 0$, there are infinitely many optima (the segment $0 \le t \le \theta_{\max}$ along that direction).

(4) Without degeneracy every pivot strictly increases z, no basis repeats, so simplex terminates.

6 Run the simplex method by hand

Method

(a) Start from the slack basis if $b \ge 0$: $s = b - Ax$, $z = 0 + c^\top x$.

(b) **Enter**: a variable with $r_j > 0$ (largest, unless told otherwise). None $\Rightarrow$ optimal, stop.

(c) **Ratio test**: only rows where the entering variable has a *negative* displayed coefficient; $\theta = \min(\text{constant}/|\text{coefficient}|)$. No such row $\Rightarrow$ unbounded.

(d) **Pivot**: solve the winning row for the entering variable, substitute into every other row and into z. Write the full new tableau and the BFS.

(e) At the end read $x^\star$ (all variables, slacks included), $z^\star$, uniqueness from the signs, y from the slack reduced costs.

Practice

Given. The running example.

Your task. Solve from the slack basis, entering the largest reduced cost. Show every tableau, ratio test, the BFS after each pivot; then give the full solution, the value, uniqueness, and the dual solution.

Solution

Tableau 0 $s_1 = 10 - 2x_1 - x_2$, $s_2 = 6 - x_1 - x_2$, $s_3 = 5 - x_2$, $z = 0 + 4x_1 + 3x_2$. BFS $(0, 0, 10, 6, 5)$.

Pivot 1 x_1 enters ($r = 4$). Ratios: s_1: $10/2 = 5$; s_2: $6/1 = 6$; s_3: x_1 absent, no limit. s_1 leaves at $\theta = 5$. Solve $x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$ and substitute:

$$x_1 = 5 - \tfrac{1}{2}x_2 - \tfrac{1}{2}s_1, \quad s_2 = 1 - \tfrac{1}{2}x_2 + \tfrac{1}{2}s_1, \quad s_3 = 5 - x_2, \quad z = 20 + x_2 - 2s_1.$$

BFS $(5, 0, 0, 1, 5)$, $z = 20$ (gain $5 \cdot 4$).

Pivot 2 x_2 enters ($r = 1$). Ratios: x_1: $5/\frac{1}{2} = 10$; s_2: $1/\frac{1}{2} = 2$; s_3: $5/1 = 5$. s_2 leaves at $\theta = 2$. Solve $x_2 = 2 + s_1 - 2s_2$:

$$x_1 = 4 - s_1 + s_2, \quad x_2 = 2 + s_1 - 2s_2, \quad s_3 = 3 - s_1 + 2s_2, \quad z = 22 - s_1 - 2s_2.$$

Read $x^\star = (x_1, x_2, s_1, s_2, s_3) = (4, 2, 0, 0, 3)$, $z^\star = 22$. Both reduced costs strictly negative $\Rightarrow$ unique. $y = (1, 2, 0)$: minus the reduced costs of s_1, s_2, and 0 for s_3 because s_3 is basic. The last row is the proof: $z = 22 - s_1 - 2s_2 \le 22$ for every feasible plan.

Gotcha

A row where the entering variable has a *positive* coefficient never limits the step (that variable grows). Forgetting the basic slack in the answer ($s_3 = 3$: resource 3 is not binding) loses the point. A zero reduced cost in an intermediate tableau says nothing about optimality yet.

7 Recognise the special cases in a tableau

Practice

Given. Four tableaux (max convention):
(i) $x_1 = 2 + s_1$, $x_2 = 3 + 2s_1$, $z = 7 + s_1$. (ii) $x_1 = 0 - s_1$, $x_2 = 4 + s_1$, $z = 8 + 2s_1$.
(iii) $x_1 = 4 - \frac{1}{2}x_2 - \frac{1}{2}s_2$, $x_3 = 2 - \frac{1}{2}x_2 - s_1 + \frac{1}{2}s_2$, $z = 14 + 0x_2 - s_1 - s_2$. (iv) $x_1 = -1 + s_1$, $x_2 = 3 - s_1$, $z = 10 - s_1$.

Your task. For each: optimal? unbounded? degenerate? unique? If optima are not unique, list them all.

Solution

(i) $r_{s_1} = 1 > 0$ and no row has a negative coefficient of s_1: **unbounded**, along $(x_1, x_2, s_1) = (2, 3, 0) + t(1, 2, 1)$.
(ii) $r_{s_1} = 2 > 0$ but the x_1 row gives $\theta_{\max} = 0/1 = 0$: **degenerate** BFS $(0, 4, 0)$; the pivot changes the basis, not the point. Not yet known to be optimal; pivot.
(iii) All $r \le 0$ and constants ≥ 0: **optimal**, $z^\star = 14$, BFS $(4, 0, 2, 0, 0)$ nondegenerate, $r_{x_2} = 0$: **infinitely many optima**. Set $x_2 = t$, $s_1 = s_2 = 0$: $x_1 = 4 - \frac{t}{2} \ge 0$, $x_3 = 2 - \frac{t}{2} \ge 0 \Rightarrow 0 \le t \le 4$. Optima: $(4 - \frac{t}{2},\ t,\ 2 - \frac{t}{2})$, from $(4, 0, 2)$ to $(2, 4, 0)$, all with $z = 14$.
(iv) Objective row fine but $x_1 = -1 < 0$: **infeasible basis**, not optimal. (A dual simplex pivot would repair it, Part VII.)

Gotcha

Check *both* conditions: constants ≥ 0 and reduced costs ≤ 0. "No limiting row" (unbounded) and "limit 0" (degenerate) are different. A zero reduced cost at a *degenerate* optimum does not by itself give a segment of optima: $\theta_{\max}$ may be 0.

Part V

Reading everything from a final tableau or a basis B

8 The dictionary: tableau $\leftrightarrow$ matrices

Method: the two directions

Direction 1, you have B (a list of basic variables in order). Compute, in this order:

1. B = those columns of A (original A with slack columns), in the stated order. c_B = their costs.
2. B^{-1} (solve $Bq = v$ or use the 2×2 formula $\frac{1}{\det}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$).
3. $x_B = B^{-1}b$ (constants). Feasible iff ≥ 0. $z_0 = c_B^\top x_B$.

4. $y^\top = c_B^\top B^{-1}$ (multipliers).
5. For each nonbasic j: $r_j = c_j - y^\top A_j$; displayed column $= -B^{-1}A_j$.
6. Optimal iff $x_B \geq 0$ and all $r_j \leq 0$.

Direction 2, you have the final tableau. Read, without computing:

you want	where it is in the equation-style tableau
basis, order	the left-hand sides, top to bottom
$x^\star$	basic = constants; every nonbasic = 0 (slacks included)
$z^\star$	the constant of the z row
binding constraints	those whose slack is nonbasic (or basic at value 0)
reduced cost r_j	coefficient of nonbasic x_j in the z row (basic: 0)
$B^{-1}A_j$	**minus** the displayed column of x_j in the basic rows
B^{-1}	**minus** the slack columns $s_1, \ldots, s_m$ side by side; a *basic* slack s_i contributes the unit column e_k where k is its row position
y_i (shadow price, dual value)	**minus** the reduced cost of s_i; 0 if s_i is basic
dual slack $t_j = (A^\top y)_j - c_j$	$-r_j$ for an original variable x_j (how far its profit is from being worth making)
basic direction of x_j	$d_B =$ its displayed column, $d_j = 1$; gain r_j per unit
$\theta_{\max}$ for x_j	ratio test over the rows with a negative coefficient of x_j
uniqueness	all nonbasic $r_j < 0$: unique; some $r_j = 0$ at a nondegenerate BFS: segment
degeneracy	a basic constant equal to 0
unboundedness	$r_j > 0$ with no negative coefficient in its column

Why the minus signs. The theorem says $x_B = B^{-1}b - B^{-1}A_N x_N$; the tableau puts the nonbasics on the right with their sign, so what you *see* is $-B^{-1}A_j$. A slack s_i has column e_i and cost 0, so its column shows $-B^{-1}e_i$ (the ith column of $-B^{-1}$) and its reduced cost is $0 - c_B^\top B^{-1}e_i = -y_i$.

Gotcha

Other books lay the tableau out differently. In the "z-row" layout (rows $x_B + B^{-1}A_N x_N = B^{-1}b$, bottom row $z + \sum \bar{c}_j x_j = z_0$ with $\bar{c}_j = y^\top A_j - c_j$), the slack columns show $+B^{-1}$, the bottom row shows $+y$ under the slacks, and "$\bar{c}_j \geq 0$ everywhere" is the optimality test. Same numbers, opposite signs. Before reading any tableau, write which layout it is. In a minimisation the improving sign flips again.

Practice

Given. The final tableau of $\max 3x_1 + 2x_2 + x_3$ s.t. $x_1 + x_2 + x_3 \leq 6$, $2x_1 + x_2 \leq 8$, $x \geq 0$ (slacks s_1, s_2):

$$x_1 = 4 - \tfrac{1}{2}x_2 - \tfrac{1}{2}s_2, \qquad x_3 = 2 - \tfrac{1}{2}x_2 - s_1 + \tfrac{1}{2}s_2, \qquad z = 14 + 0\,x_2 - s_1 - s_2.$$

Your task. Read off: (a) the ordered basis and the full $x^\star$; (b) $z^\star$ and which constraints bind; (c) B, B^{-1}, and check $BB^{-1} = I$; (d) y and verify it is dual feasible with $b^\top y = z^\star$; (e) the dual slack of x_2 and what it means; (f) $B^{-1}A_2$ for the column of x_2; (g) is the optimum unique? list all optima; (h) the shadow price and range of resource 1; (i) the range of resource 2; (j) how much the profit of x_3 may change; (k) a new product with column $(1, 1)$ and profit $\frac{5}{2}$: make it or not, and the new optimum if yes.

Solution

(a) Basis (x_1, x_3) in that order; nonbasic x_2, s_1, s_2. $x^\star = (x_1, x_2, x_3, s_1, s_2) = (4, 0, 2, 0, 0)$.

(b) $z^\star = 14$. Both slacks nonbasic (zero): both constraints bind ($4 + 0 + 2 = 6$, $8 + 0 = 8$).

(c) $B = [A_1\ A_3] = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$. Slack columns displayed: $s_1 \to (0, -1)^\top$, $s_2 \to (-\frac{1}{2}, \frac{1}{2})^\top$, so $-B^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{bmatrix}$. Check: $\begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Also $B^{-1}b = (4, 2)^\top = x_B$.

(d) $y = (1, 1)$, minus the slack reduced costs; or $c_B^\top B^{-1} = (3, 1)B^{-1} = (1, 1)$. Dual: $\min 6y_1 + 8y_2$ s.t. $y_1 + 2y_2 \geq 3$, $y_1 + y_2 \geq 2$, $y_1 \geq 1$, $y \geq 0$. Check: $3 \geq 3$, $2 \geq 2$, $1 \geq 1$; $b^\top y = 14 = z^\star$. Optimal pair.

(e) $t_2 = -r_2 = 0$: the dual row of x_2 is tight ($y_1 + y_2 = 2 = c_2$). Product 2 exactly breaks even: making it neither gains nor loses.

(f) $B^{-1}A_2 = -$(displayed column of x_2) $= (\frac{1}{2}, \frac{1}{2})^\top$. Check: $B(\frac{1}{2}, \frac{1}{2})^\top = (1, 1)^\top = A_2$.

(g) Not unique: $r_{x_2} = 0$ at a nondegenerate BFS. $x_2 = t$, $s_1 = s_2 = 0$: $x_1 = 4 - \frac{t}{2}$, $x_3 = 2 - \frac{t}{2}$, so $0 \leq t \leq 4$: optima $(4 - \frac{t}{2},\ t,\ 2 - \frac{t}{2})$, from $(4, 0, 2)$ to $(2, 4, 0)$, value 14.

(h) $y_1 = 1$. $g = B^{-1}e_1 = (0, 1)^\top$ (minus the s_1 column): new constants $x_1 = 4$, $x_3 = 2 + \delta \geq 0 \Rightarrow \delta \geq -2$, no upper limit. Range $b_1 \in [4, \infty)$, $z^\star = 14 + \delta$ there.

(i) $g = B^{-1}e_2 = (\frac{1}{2}, -\frac{1}{2})^\top$: $4 + \frac{\delta}{2} \geq 0$, $2 - \frac{\delta}{2} \geq 0 \Rightarrow -8 \leq \delta \leq 4$, slope $y_2 = 1$.

(j) $c_3 = 1 + \varepsilon$, x_3 basic: add $\varepsilon \times$(x_3 row) to the z row: $z = 14 + 2\varepsilon + (-\frac{\varepsilon}{2})x_2 + (-1 - \varepsilon)s_1 + (-1 + \frac{\varepsilon}{2})s_2$. All ≤ 0: $\varepsilon \geq 0$, $\varepsilon \geq -1$, $\varepsilon \leq 2$. Range $0 \leq \varepsilon \leq 2$: the lower end is 0 because $r_{x_2} = 0$ already; any decrease makes x_2 strictly better.

(k) $r_u = \frac{5}{2} - y^\top a = \frac{5}{2} - 2 = \frac{1}{2} > 0$: make it. Column $-B^{-1}a = -(\frac{1}{2}, \frac{1}{2})^\top$: $x_1 = 4 - \frac{u}{2}$, $x_3 = 2 - \frac{u}{2}$; x_3 leaves at $u = 4$. Pivot: $u = 4 - x_2 - 2x_3 - 2s_1 + s_2$, $x_1 = 2 + x_3 + s_1 - s_2$, $z = 16 - \frac{1}{2}x_2 - x_3 - 2s_1 - \frac{1}{2}s_2$. New optimum $(x_1, x_2, x_3, u) = (2, 0, 0, 4)$, $z^\star = 16$, unique, $y = (2, \frac{1}{2})$.

Gotcha

Rows must be read in the *stated basic order*: B^{-1}'s rows follow the rows of the tableau, its columns follow $s_1, \ldots, s_m$. When a slack is basic its column is not displayed: insert a unit vector. $t_j = -r_j$ only for original variables; for a slack, $-r_{s_i}$ is y_i itself. The range of a basic cost can have 0 as an endpoint (part j): that is not an error, it is a tie.

9 From B to the whole tableau (direction 1, worked)

Practice

Given. The running example, basis (x_1, x_2, s_3) in that order.

Your task. Compute B, B^{-1}, x_B, z_0, y, every reduced cost and the displayed columns, decide whether the basis is optimal, and write the tableau.

Solution

$B = [A_{x_1}\ A_{x_2}\ A_{s_3}] = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $c_B = (4, 3, 0)$. Solve $Bq = v$: $2q_1 + q_2 = v_1$, $q_1 + q_2 = v_2$, $q_2 + q_3 = v_3$ give $q_1 = v_1 - v_2$, $q_2 = 2v_2 - v_1$, $q_3 = v_3 - 2v_2 + v_1$, hence

$$B^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 1 & -2 & 1 \end{bmatrix}, \quad x_B = B^{-1}(10, 6, 5)^\top = (4, 2, 3)^\top \geq 0, \quad z_0 = 4 \cdot 4 + 3 \cdot 2 = 22.$$

$y^\top = c_B^\top B^{-1} = (4 - 3,\ -4 + 6,\ 0) = (1, 2, 0)$. Nonbasic s_1 (column e_1, cost 0): $r = 0 - y_1 = -1$, displayed column $-B^{-1}e_1 = (-1, 1, -1)^\top$. Nonbasic s_2: $r = -y_2 = -2$, column $-B^{-1}e_2 = (1, -2, 2)^\top$. Both $r \leq 0$ and $x_B \geq 0$: optimal. Tableau: $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$, the same as the one simplex reached.

Gotcha

B is built from the *original* A (with slack columns), never from a later tableau. A basic slack (s_3) has $r = 0$ and $y_3 = 0$ by complementary slackness: its resource is not binding ($s_3 = 3$).

Part VI

Station 4: Certify (duality)

10 The theorems

Theorem: Canonical dual; weak and strong duality

Primal $\max\{c^\top x : Ax \le b, x \ge 0\}$, dual $\min\{b^\top y : A^\top y \ge c, y \ge 0\}$. One primal row $\to$ one dual variable; one primal variable $\to$ one dual row.

Weak duality. Primal-feasible x, dual-feasible y: $c^\top x \le y^\top Ax \le b^\top y$. So a feasible pair with $c^\top x = b^\top y$ is optimal on both sides (**matching pair**).

Strong duality. If the primal is feasible with bounded objective, both have optimal solutions and the values are equal. The optimal y is $c_B^\top B^{-1}$ of an optimal basis (read off the final tableau).

Proof idea. Weak: $c \le A^\top y$ and $x \ge 0$ give $c^\top x \le (A^\top y)^\top x = y^\top Ax$; $Ax \le b$ and $y \ge 0$ give $y^\top Ax \le y^\top b$. Strong: the final tableau's $r \le 0$ on original columns says $A^\top y \ge c$, on slack columns says $y \ge 0$, and $b^\top y = c_B^\top B^{-1} b = z^\star$.

Theorem: Dualization rules, max primal / min dual

primal row	dual variable	primal variable	dual row
$a_i^\top x \le b_i$	$y_i \ge 0$	$x_j \ge 0$	$A_j^\top y \ge c_j$
$a_i^\top x \ge b_i$	$y_i \le 0$	$x_j \le 0$	$A_j^\top y \le c_j$
$a_i^\top x = b_i$	$y_i \in \mathbb{R}$	$x_j \in \mathbb{R}$	$A_j^\top y = c_j$

Dual objective $\min b^\top y$. The dual of the dual is the primal. Reason: choose the sign of y_i so that $y_i a_i^\top x \le y_i b_i$ stays true; choose the row direction so that $c_j x_j \le (A_j^\top y) x_j$ for the allowed sign of x_j.

Theorem: Complementary slackness; the outcome table

Primal slack $s = b - Ax$, dual slack $t = A^\top y - c$. For feasible x, y: $b^\top y - c^\top x = \sum_i y_i s_i + \sum_j x_j t_j$, every term ≥ 0. Hence x, y are both optimal iff $y_i s_i = 0\ \forall i$ and $x_j t_j = 0\ \forall j$: a positive dual price needs a tight row; a positive activity needs a tight dual row. Tightness alone forces nothing.
Outcomes. One side finite $\Rightarrow$ both finite and equal. Primal unbounded $\Rightarrow$ dual infeasible. Dual unbounded $\Rightarrow$ primal infeasible. Primal infeasible $\Rightarrow$ dual unbounded *or* infeasible (both can be infeasible).

11 Write a dual with mixed signs

Method

Rule 0: get a *max* primal (negate a min objective). Then one row of the table per primal row, one per primal variable; dual objective $b^\top y$ with the *original* right-hand sides; coefficients of dual row j are read *down* column j of A. Alternative: convert the primal to canonical form first and dualise that; both give equivalent duals.

Practice

Given. $\max 2x_1 - x_2 + 3x_3$ s.t. $x_1 + x_2 + x_3 \le 10$, $x_1 - x_3 \ge 2$, $2x_1 + x_2 = 6$, $x_1 \ge 0$, $x_2 \le 0$, $x_3 \in \mathbb{R}$.

Your task. Write the dual and justify every sign and direction.

Solution

Rows: $\le$ gives $y_1 \ge 0$; $\ge$ gives $y_2 \le 0$; $=$ gives $y_3 \in \mathbb{R}$. Columns: $x_1 \ge 0$: $y_1 + y_2 + 2y_3 \ge 2$; $x_2 \le 0$: $y_1 + 0y_2 + y_3 \le -1$; $x_3 \in \mathbb{R}$: $y_1 - y_2 + 0y_3 = 3$. Dual: $\min 10y_1 + 2y_2 + 6y_3$ subject to those three rows and $y_1 \ge 0$, $y_2 \le 0$, y_3 free.

Gotcha

A $\ge$ row in a max primal gives a *nonpositive* dual variable (multiplying by $y_i \le 0$ turns $\ge$ into $\le$). The dual row of a free variable is an equality; the dual variable of an equality row is free. The dual objective keeps b as written, even $b_2 = 2$ from the $\ge$ row.

12 Certify a candidate with complementary slackness

Method

(a) Check the candidate $\hat{x}$ is primal feasible (every row, every sign). If not, stop: not optimal.

(b) Positive $\hat{x}_j \Rightarrow$ dual row j is an equation. Slack row i ($s_i > 0$) $\Rightarrow y_i = 0$.

(c) Solve the resulting linear system for y.

(d) **Verify** y: signs and the *remaining* dual rows. Then $b^\top y = c^\top \hat{x}$ closes the proof. If the system has no feasible y, $\hat{x}$ is not optimal.

Practice

Given. The running example. Candidates $\hat{x} = (4, 2)$ and $\tilde{x} = (5, 0)$.

Your task. Decide optimality of each by complementary slackness.

Solution

Dual: $\min 10y_1 + 6y_2 + 5y_3$ s.t. $2y_1 + y_2 \geq 4$, $y_1 + y_2 + y_3 \geq 3$, $y \geq 0$.
$\hat{x} = (4, 2)$: feasible ($10 \leq 10$, $6 \leq 6$, $2 \leq 5$). Row 3 slack ($s_3 = 3$) $\Rightarrow y_3 = 0$. $\hat{x}_1, \hat{x}_2 > 0 \Rightarrow 2y_1 + y_2 = 4$, $y_1 + y_2 = 3 \Rightarrow y = (1, 2, 0)$. Verify: $y \geq 0$, both rows tight; $b^\top y = 10 + 12 = 22 = c^\top \hat{x}$. Optimal.
$\tilde{x} = (5, 0)$: feasible ($10 \leq 10$, $5 \leq 6$, $0 \leq 5$). Rows 2, 3 slack $\Rightarrow y_2 = y_3 = 0$. $\tilde{x}_1 > 0 \Rightarrow 2y_1 = 4$, $y_1 = 2$. Verify the dual row of x_2: $y_1 + y_2 + y_3 = 2 \geq 3$ fails. No certificate exists, so $\tilde{x}$ is not optimal (indeed $z = 20 < 22$).

Gotcha

Complementary slackness *finds* the candidate y; only feasibility of both plus equal values *proves* optimality. A tight primal row does not force $y_i > 0$ (both can be zero). Applying the conditions to an infeasible $\hat{x}$ proves nothing.

13 Classify a primal–dual pair

Practice

Given. (a) $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 1$, $x \geq 0$. (b) $\max x_1$ s.t. $x_1 \leq 1$, $x_1 \geq 2$, $x_1 \geq 0$.

Your task. Classify each primal and its dual, with proof.

Solution

(a) Ray (t, t) is feasible for all $t \geq 0$ ($t - t = 0 \leq 1$) with $z = 2t \to \infty$: **unbounded**. Dual: $\min y$ s.t. $y \geq 1$, $-y \geq 1$, $y \geq 0$: the second row forces $y \leq -1$: **infeasible**, as the theorem predicts. (b) $x_1 \leq 1$ and $x_1 \geq 2$: **infeasible**. Dual (max primal, rows $\leq$ and $\geq$): $\min y_1 + 2y_2$ s.t. $y_1 + y_2 \geq 1$, $y_1 \geq 0$, $y_2 \leq 0$: take $y_1 = 1 + k$, $y_2 = -k$... value $1 + k - 2k = 1 - k \to -\infty$: **unbounded**. The table allows this (infeasible primal, unbounded dual).

Gotcha

"Primal infeasible $\Rightarrow$ dual unbounded" is **false** as a rule (both may be infeasible: $\max x_1$ s.t. $x_1 - x_2 \leq -1$, $-x_1 + x_2 \leq -1$). An unbounded feasible *region* does not mean an unbounded *objective* ($\max -x_1$, $x_1 \geq 0$ has optimum 0).

Part VII

Station 5: Update (sensitivity analysis)

14 The vocabulary: what every symbol means and where to read it

Sensitivity analysis asks: *the optimal tableau is in front of me; the data change a little; what can I say without solving again?* Everything below is about one fixed optimal basis B and the quantities attached to it. The lecture writes $v_{\max}(\delta)$ for the optimal value as a function of the changed datum and $p^\top = c_B^\top B^{-1}$ for the multipliers; this guide writes $z^\star(\delta)$ and y.

symbol	meaning	where it is in the final tableau
B, N	basic / nonbasic index sets, $\lvert B\rvert = m$	basics are the left-hand variables, nonbasics the right-hand ones
B^{-1}	inverse of the basis matrix	minus the slack columns (a basic slack contributes a unit column)
$x_B = B^{-1}b$	the **plan**: production levels and leftover slacks	the constant column β
$z_0 = c_B^\top x_B$	the optimal value $z^\star$	the constant of the z row
$y^\top = c_B^\top B^{-1}$	**multipliers**, shadow prices, dual optimum: value of one more unit of each resource	$y_i = -r_{s_i}$, minus the z-row coefficient of slack s_i
$r_j = c_j - y^\top A_j$	**reduced cost**: profit of x_j minus the shadow value of what one unit of x_j consumes	the z-row coefficient of x_j (0 for basics)
$Q = -B^{-1}A_N$	the displayed body of the tableau	coefficient of nonbasic x_j in row k is q_{kj}
$g = B^{-1}e_i$	**response vector**: how the plan moves per unit of resource i	minus the column of s_i (so $g = e_k$ if s_i is the k-th basic)
valid range	interval of a single change over which B stays optimal	computed from the two tests, always contains 0
local formula	$z^\star(\delta) = z^\star + y_i\delta$, or $z^\star(\varepsilon) = z^\star + \varepsilon x_{B(k)}$	valid on the range only
break-even	change that makes a reduced cost reach 0	$-r_j$ for a nonbasic profit; $y^\top a$ for a new activity
dual simplex pivot	pivot that repairs a negative constant while keeping $r \le 0$	used when a resource change leaves the range

15 The theorems, with proofs

Theorem: T1. The two tests, and which data each depends on

With the data (A, b, c) fixed and B a basis, define $x_B = B^{-1}b$, $y^\top = c_B^\top B^{-1}$, $r^\top = c^\top - y^\top A$. Then

$$B \text{ is an optimal basis} \iff \underbrace{B^{-1}b \ge 0}_{\text{feasibility test}} \quad \text{and} \quad \underbrace{r \le 0}_{\text{optimality test}} .$$

Moreover B^{-1} depends only on A; x_B depends on (A, b) but not on c; y and r depend on (A, c) but not on b.

Proof idea. ($\Leftarrow$) Any feasible x satisfies $Ax = b$, so $x_B = B^{-1}b - B^{-1}A_N x_N$ and $c^\top x = c_B^\top x_B + c_N^\top x_N = z_0 + r_N^\top x_N \le z_0$ because $r_N \le 0$ and $x_N \ge 0$. The BFS of B attains z_0, so it is optimal. ($\Rightarrow$, non-degenerate case) if $r_j > 0$ the basic direction of x_j keeps feasibility for a small step and increases z. The dependence statements are read off the formulas.

Used for. This is the whole logic of sensitivity: a change in b can only break the feasibility test, a change in c only the optimality test, a new column only adds one new reduced cost. That is the table of the next box.

Theorem: T2. Which test each change can break

change	unchanged	can fail	repair if it fails
$b \to b'$	B^{-1}, y, all r_j, all displayed columns	$B^{-1}b' \geq 0$	dual simplex pivot(s)
$c_j \to c'_j$, j nonbasic	everything except r_j	$r'_j \leq 0$	ordinary pivot, x_j enters
$c_{B(k)} \to c'_{B(k)}$	the plan x_B, the body Q	every r_j, and y moves	ordinary pivot
new column a, profit γ	everything old	$r_u = \gamma - y^\top a \leq 0$	ordinary pivot, u enters

Proof idea. Immediate from the dependence statements of T1.

Theorem: T3. The value identity $z^\star = y^\top b$

For any basis, $z_0 = c_B^\top B^{-1} b = y^\top b$. In words: the optimal value equals the resources priced at their multipliers.

Proof idea. Associativity: $c_B^\top (B^{-1} b) = (c_B^\top B^{-1}) b$.

Used for. This one line is where every "slope" result comes from: when b moves and y does not, z_0 moves linearly with slope y.

Theorem: T4. Resource change: response, valid range, shadow price

Let $b' = b + \delta e_i$ and $g = B^{-1} e_i$. Then for every δ:

(i) the new constants are $x_B(\delta) = x_B + \delta g$, and nothing else in the tableau changes;

(ii) B stays optimal iff $x_B + \delta g \geq 0$, i.e.

$$\delta_{\min} := \max_{k:\, g_k > 0} \Big(- \frac{x_{B(k)}}{g_k} \Big) \;\leq\; \delta \;\leq\; \min_{k:\, g_k < 0} \frac{x_{B(k)}}{-g_k} =: \delta_{\max}$$

(an empty max is $-\infty$, an empty min is $+\infty$; the range contains 0);

(iii) on that range $z^\star(\delta) = z^\star + y_i\, \delta$: the multiplier y_i is the **shadow price** of resource i, the rate at which the optimal value grows per extra unit, and the optimal *plan* is $x_B + \delta g$, which does move;

(iv) if s_i is basic (resource i not binding) then $y_i = 0$ and the range is $[-x_{s_i}, +\infty)$.

Proof idea. (i) $B^{-1} b' = B^{-1} b + \delta B^{-1} e_i$ by linearity; B^{-1}, y, r, Q do not involve b (T1). The column of s_i in A is e_i, so $g = B^{-1} A_{s_i}$ is minus the displayed column of s_i. (ii) Row k requires $x_{B(k)} + \delta g_k \geq 0$: if $g_k > 0$ this says $\delta \geq -x_{B(k)}/g_k$, if $g_k < 0$ it says $\delta \leq x_{B(k)}/(-g_k)$, if $g_k = 0$ nothing. Intersect. $\delta = 0$ satisfies all of them because $x_B \geq 0$. (iii) The optimality test is untouched, so B is optimal and T3 gives $z^\star(\delta) = y^\top (b + \delta e_i) = z^\star + \delta y_i$. (iv) If $s_i = x_{B(k)}$ then $B e_k = e_i$, so $g = B^{-1} e_i = e_k$ and $y_i = c_B^\top e_k = c_{s_i} = 0$; the only constraint is $x_{s_i} + \delta \geq 0$.

Used for. Price of an extra hour, how far a price is valid, how the plan reacts. The sign of y_i for a $\leq$ constraint in a max problem is ≥ 0 because $y_i = -r_{s_i}$ and $r_{s_i} \leq 0$ at an optimal tableau: more of a resource never hurts.

Theorem: T5. The value function in b and the two global bounds

Let $z^\star(b)$ be the optimal value as a function of the right-hand side, on the set of b where the LP is feasible and bounded. Then

(i) $z^\star$ is **concave and piecewise linear** in b; on each piece its gradient is the optimal multiplier vector y of a basis;

(ii) (upper bound, always) $z^\star(b+e) \le z^\star(b) + e^\top y^\star$ for every change e, where $y^\star$ is an optimal dual solution at b;

(iii) (lower bound, always) $z^\star(b+e) \ge z^\star(b) + e^\top \tilde{y}$ where $\tilde{y}$ is an optimal dual solution at $b+e$.

Proof idea. By strong duality $z^\star(b) = \min\{b^\top y : A^\top y \ge c,\ y \ge 0\}$. The dual feasible set does not depend on b, so $z^\star(b)$ is the minimum of the linear functions $b \mapsto b^\top y$ over a fixed set (its finitely many vertices suffice): a minimum of linear functions is concave and piecewise linear. (ii) $y^\star$ is still dual feasible at $b+e$, so by weak duality $z^\star(b+e) \le (b+e)^\top y^\star = z^\star(b) + e^\top y^\star$. (iii) Symmetric: $\tilde{y}$ is dual feasible at b, so $z^\star(b) \le b^\top \tilde{y} = z^\star(b+e) - e^\top \tilde{y}$.

Used for. Explains why the local formula $z^\star + y_i\delta$ *overestimates* the value beyond the range when $\delta > 0$ (concavity: the slope can only decrease), why resources have decreasing marginal value, and how to bound the optimal value after a change without recomputing. The midterm question "prove $v_0 - \sigma \le v \le v_0 + \sigma$" is (ii)+(iii) with $|e^\top y| \le \sigma$, or its twin for the objective (T8).

Theorem: T6. Dual simplex pivot: what it does and why it is safe

Suppose the tableau has all $r_j \le 0$ but a constant $\beta_\ell < 0$ in row ℓ (equation $x_{B(\ell)} = \beta_\ell + \sum_j q_{\ell j} x_j$).

(i) If every $q_{\ell j} \le 0$, the LP is **infeasible**.

(ii) Otherwise choose the entering x_j among $\{j : q_{\ell j} > 0\}$ minimising $(-r_j)/q_{\ell j}$; solve row ℓ for x_j and substitute everywhere. The new tableau still has all reduced costs ≤ 0, and its constant z_0 does not increase.

(iii) Repeating until all constants are ≥ 0 yields an optimal tableau (or proves infeasibility). Only finitely many pivots are needed if ties are broken by a fixed rule.

Proof idea. (i) For every $x_N \ge 0$, $x_{B(\ell)} = \beta_\ell + \sum_j q_{\ell j} x_j \le \beta_\ell < 0$: the row cannot be made nonnegative. (ii) Solving the row: $x_j = \frac{-\beta_\ell}{q_{\ell j}} + \frac{1}{q_{\ell j}} x_{B(\ell)} - \sum_{k \ne j} \frac{q_{\ell k}}{q_{\ell j}} x_k$; the new constant $-\beta_\ell/q_{\ell j} > 0$. Substituting in $z = z_0 + \sum r_k x_k$: the coefficient of the leaving $x_{B(\ell)}$ becomes $r_j/q_{\ell j} \le 0$; the coefficient of another nonbasic x_k becomes $r_k - r_j\, q_{\ell k}/q_{\ell j}$. If $q_{\ell k} \le 0$ this is $\le r_k \le 0$ since $-r_j q_{\ell k}/q_{\ell j} \le 0$. If $q_{\ell k} > 0$ it is ≤ 0 iff $(-r_k)/q_{\ell k} \ge (-r_j)/q_{\ell j}$, which is exactly the minimum-ratio choice. The new constant of z is $z_0 + r_j(-\beta_\ell)/q_{\ell j} = z_0 - r_j\beta_\ell/q_{\ell j}$, and $r_j \le 0$, $\beta_\ell < 0$, $q_{\ell j} > 0$ make the correction ≤ 0. (iii) Each tableau is dual feasible ($r \le 0$), so the first one that is also primal feasible is optimal by T1.

Used for. Reoptimising after a resource change leaves the range. Reading: the dual simplex walks down the dual objective (z_0 decreases) while keeping dual feasibility; the ordinary simplex walks up the primal objective while keeping primal feasibility.

Theorem: T7. Profit (objective) change

(i) **Nonbasic** $c_j \to c_j + t$: only $r_j \to r_j + t$ changes. B stays optimal iff $t \le -r_j$. The quantity $-r_j \ge 0$ is the **break-even increase**: how much more profitable x_j must become before it is worth producing. Plan and value are unchanged on the range.

(ii) **Basic** $c_{B(k)} \to c_{B(k)} + \varepsilon$: the body and the constants do not move; the z row becomes $z_0 + \varepsilon\beta_k + \sum_j (r_j + \varepsilon q_{kj}) x_j$, i.e. "add ε times the row of $x_{B(k)}$ to the z row". Equivalently

$y^\top \to y^\top + \varepsilon\,(B^{-1})_{k\cdot}$ (row k of B^{-1}). B stays optimal iff $r_j + \varepsilon q_{kj} \le 0$ for all nonbasic j:

$$\max_{j:\, q_{kj}<0} \frac{-r_j}{q_{kj}} \;\le\; \varepsilon \;\le\; \min_{j:\, q_{kj}>0} \frac{-r_j}{q_{kj}}.$$

On that range the *plan* $x_B = B^{-1}b$ is unchanged and the value is $z^\star(\varepsilon) = z^\star + \varepsilon\, x_{B(k)}$: the slope is the quantity produced.

(iii) The value function $c \mapsto z^\star(c)$ is **convex and piecewise linear**; on each piece its gradient is the optimal plan $x^\star$.

Proof idea. (i) $y = c_B^\top B^{-1}$ does not involve c_j, so $r_j = c_j - y^\top A_j$ is the only reduced cost that moves, by exactly t. (ii) Write the objective as $c^\top x + \varepsilon x_{B(k)}$ and substitute the row $x_{B(k)} = \beta_k + \sum_j q_{kj} x_j$ of the tableau: that adds $\varepsilon\beta_k$ to the constant and εq_{kj} to each r_j. The range is the intersection of the inequalities $r_j + \varepsilon q_{kj} \le 0$ solved for ε according to the sign of q_{kj}; $\varepsilon = 0$ satisfies them. Feasibility $B^{-1}b \ge 0$ does not involve c. (iii) $z^\star(c) = \max\{c^\top x : x \in P\} = \max$ over the finitely many vertices of P of the linear functions $c \mapsto c^\top x^{(v)}$: a maximum of linear functions is convex piecewise linear, with gradient the maximising vertex.

Used for. Pricing decisions; "how much must the profit of x_j rise before we make it"; checking that a plan is robust to profit estimates. Note the duality of slopes: in b the slope is the dual solution y, in c it is the primal solution $x^\star$.

Theorem: T8. Stability of the optimal value under an objective perturbation

Let $P = \{x : Ax \le b,\ x \ge 0\}$ be bounded, $v_0 = \max_{x\in P} c^\top x$ and $v = \max_{x\in P}(c+d)^\top x$ for a perturbation d. If $|d^\top x| \le \sigma$ for every $x \in P$ (for instance $|d_j| \le \eta$ for all j and $\sum_j x_j \le M$ on P, giving $\sigma = \eta M$), then

$$v_0 - \sigma \;\le\; v \;\le\; v_0 + \sigma.$$

Proof idea. Let $x^\star$ be optimal for c and $\tilde{x}$ optimal for $c+d$; both lie in P because the constraints did not change. Lower bound: $v \ge (c+d)^\top x^\star = v_0 + d^\top x^\star \ge v_0 - \sigma$. Upper bound: $v = c^\top \tilde{x} + d^\top \tilde{x} \le v_0 + \sigma$ because $c^\top \tilde{x} \le v_0$. The trick is "the optimum of one problem is a feasible candidate for the other", used once in each direction.

Used for. Any exam question of the form "show the true optimal value lies within σ of the computed one". For a right-hand-side perturbation use T5(ii)(iii) instead: $x^\star$ is no longer feasible, but $y^\star$ is still dual feasible.

Theorem: T9. New activity (new column)

Add a variable $u \ge 0$ with column $a \in \mathbb{R}^m$ and profit γ. Keep the old basis B. Then

(i) the old tableau is still the tableau of B for the enlarged problem, with one new column: $x_B = \beta - (B^{-1}a)\,u + \ldots$, i.e. displayed column $-B^{-1}a$, and z-row coefficient $r_u = \gamma - y^\top a$;

(ii) if $r_u \le 0$ the old plan with $u = 0$ is optimal for the enlarged problem; the **break-even profit** is $\gamma_0 = y^\top a$, the resources one unit of u consumes priced at the shadow prices (its *opportunity cost*);

(iii) if $r_u > 0$, u enters: ratio test on the column $-B^{-1}a$, pivot, continue the simplex. The new optimal value is at least the old one.

Proof idea. (i) $Ax + au = b$ gives $x_B = B^{-1}b - B^{-1}A_N x_N - B^{-1}a\,u$ and $z = c_B^\top x_B + c_N^\top x_N + \gamma u = z_0 + r_N^\top x_N + (\gamma - c_B^\top B^{-1}a)u$. (ii) T1 applied to the enlarged tableau: feasibility is the old $\beta \ge 0$, optimality is the old $r \le 0$ plus $r_u \le 0$. (iii) The old BFS is feasible for the enlarged LP, so its optimum is $\ge z_0$.

Used for. "Should we launch product u?" Compare γ with $y^\top a$. No new row, no new slack, B

stays $m \times m$.

Theorem: T10. New constraint

Add a constraint $\alpha^\top x \le \beta$. If the old optimum $x^\star$ satisfies it, $x^\star$ stays optimal (the feasible set shrank and still contains the old best point). If not, add a slack row $s_{m+1} = \beta - \alpha^\top x$, rewrite it in terms of the nonbasics by substituting the tableau rows; its constant is $\beta - \alpha^\top x^\star < 0$ and all reduced costs are still ≤ 0: run the dual simplex (T6). The new value is $\le$ the old one.

Proof idea. Shrinking the feasible set cannot raise a maximum; the substituted row is exactly a tableau row with a negative constant and the z row is untouched.

Theorem: T11. Simultaneous changes

If $b \to b + \delta e_i$ and $c_{B(k)} \to c_{B(k)} + \varepsilon$ at the same time, B stays optimal iff δ lies in its own range (feasibility test) **and** ε lies in its own range (optimality test): the set of admissible (δ, ε) is the rectangle. The value is

$$z^\star(\delta, \varepsilon) = c_B(\varepsilon)^\top B^{-1} b(\delta) = z^\star + y_i \delta + x_{B(k)} \varepsilon + \varepsilon g_k \delta,$$

not the sum of the separate changes: the cross term $\varepsilon g_k \delta$ is the extra profit on the extra quantity. Two changes of the *same* kind (two resources, or two profits) are *not* independent: the inequalities must be rewritten with both parameters, $x_B + \delta g + \delta' g' \ge 0$, which gives a polygon, not a rectangle.

Proof idea. T1: the feasibility test involves only b, the optimality test only c; so each test is a condition on one parameter alone. For the value, expand $(c_B + \varepsilon e_k)^\top (x_B + \delta g)$.

Gotcha

Four confusions that cost points. (1) "Same basis" is not "same plan": after a resource change the plan is $x_B + \delta g$; after a profit change the plan is unchanged but y and all reduced costs move. (2) The shadow price is a slope valid on the range only; outside it concavity says the real gain is smaller. (3) g is *minus* the displayed column of s_i in this equation-style layout; in a z-row layout the slack columns show $+B^{-1}$ and the sign flips. (4) A range computed from the tableau is for *one* change at a time, with everything else fixed.

16 The method: sensitivity at exam speed

Method

(a) **Name the change** and look it up in T2: resource (feasibility test), profit of a nonbasic (one reduced cost), profit of a basic (whole z row), new activity (one new reduced cost), new constraint (one new row).

(b) **Read what you need** from the tableau: x_B (constants), y (minus slack reduced costs), g (minus the slack column), the row of the basic variable whose profit changes, or B^{-1} (minus the slack columns) for a new activity.

(c) **Write the test as inequalities** in the parameter, one per row (resource) or one per nonbasic column (profit). Solve each for the parameter, watching the sign you divide by. Intersect. Check that 0 is inside.

(d) **Inside the range**, state plan and value with the local formula: resource, plan $x_B + \delta g$ and value $z^\star + y_i \delta$; profit, plan unchanged and value $z^\star + \varepsilon x_{B(k)}$ (or unchanged for a nonbasic); new activity with $r_u \le 0$, nothing changes.

(e) **Outside the range**, write the tableau with the new numbers and pivot: a negative constant

means dual simplex (T6), a positive reduced cost means ordinary simplex. Read the new plan, value and y, and say which formula stopped being valid.

(f) **Sanity checks**: the new plan satisfies the original constraints; the value identity $z^\star = y^\top b$ holds; $y \geq 0$ with $y_i = 0$ for every non-binding resource.

Practice

Given. The running example and its optimal tableau $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$. The profit estimates are uncertain: each true profit lies within 0.5 of the estimated 4 and 3.

Your task. (a) Prove that the true optimal value v satisfies $19 \leq v \leq 25$. (b) Use the tableau to show that the *plan* $(4, 2)$ stays optimal for every such profit vector. (c) Deduce the exact smallest and largest possible values of v.

Solution

Bound (T8). Here $d = (d_1, d_2)$ with $|d_j| \leq \frac{1}{2}$. On the feasible set $x_1 + x_2 \leq 6$, so $|d^\top x| \leq \frac{1}{2}(x_1 + x_2) \leq 3 =: \sigma$. With $v_0 = 22$: $19 \leq v \leq 25$.

Plan stays optimal (T7). Add $\varepsilon_1 \times (x_1 \text{ row}) + \varepsilon_2 \times (x_2 \text{ row})$ to the z row: coefficient of s_1 is $-1 - \varepsilon_1 + \varepsilon_2$, of s_2 is $-2 + \varepsilon_1 - 2\varepsilon_2$. With $|\varepsilon_j| \leq \frac{1}{2}$: $-1 - \varepsilon_1 + \varepsilon_2 \leq -1 + \frac{1}{2} + \frac{1}{2} = 0$ and $-2 + \varepsilon_1 - 2\varepsilon_2 \leq -2 + \frac{1}{2} + 1 \leq 0$. Both tests hold (feasibility did not change), so $(4, 2)$ is optimal for every such c.

Exact extremes. $v = (4 + \varepsilon_1)4 + (3 + \varepsilon_2)2 = 22 + 4\varepsilon_1 + 2\varepsilon_2 \in [22 - 3,\ 22 + 3] = [19, 25]$. The T8 bound is attained here, because the plan does not move and $x_1 + x_2 = 6$ at the optimum; in general T8 is only a bound.

Gotcha

In T8 the feasible set must be bounded, otherwise σ does not exist; say so in the proof. Do not claim $x^\star$ stays optimal in general: you only need it to stay *feasible*, which is free when only c changes. For a change in b switch to the dual (T5): $x^\star$ may become infeasible, but $y^\star$ remains dual feasible.

17 Resource change: range, value, and beyond the range

Method

(a) Read $g = B^{-1}e_i$ = minus the column of s_i (unit vector if s_i is basic, then the range is one-sided and $y_i = 0$).

(b) One inequality per basic row: constant $+\delta g_k \geq 0$. Intersect. The range always contains 0.

(c) Inside: plan $x_B + \delta g$ (it moves!), value $z^\star + y_i\delta$.

(d) Outside: write the tableau with the new constants, find the negative one, dual simplex pivot, read the new plan, value and y.

Practice

Given. The optimal tableau of the running example: $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$. Resource 1 becomes $10 + \delta$.

Your task. (a) Range of δ and $z^\star(\delta)$. (b) Plan and value for $\delta = 1$. (c) $\delta = 4$: show the basis fails, reoptimise, give plan, value, y, and compare with the local formula. (d) Range and shadow price of resource 2 and of resource 3.

Solution

(a) $g = B^{-1}e_1 = -(-1,1,-1)^\top = (1,-1,1)^\top$. $4+\delta \geq 0$, $2-\delta \geq 0$, $3+\delta \geq 0 \Rightarrow -3 \leq \delta \leq 2$. $z^\star = 22+\delta$ ($y_1 = 1$).

(b) $\delta = 1$: $(x_1, x_2, s_3) = (5,1,4)$, $z = 23$. Same basis, different plan.

(c) $\delta = 4$: constants $(8,-2,7)$: $x_2 < 0$, infeasible. Row $x_2 = -2 + s_1 - 2s_2$; positive coefficient only on s_1 (ratio $-r_{s_1}/1 = 1$): s_1 enters, x_2 leaves. $s_1 = 2 + x_2 + 2s_2$; substitute: $x_1 = 8 - s_1 + s_2 = 6 - x_2 - s_2$, $s_3 = 7 - s_1 + 2s_2 = 5 - x_2$, $z = 26 - s_1 - 2s_2 = 24 - x_2 - 4s_2$. New optimum $(6,0)$, $z^\star = 24$, $y = (0,4,0)$: resource 1 is no longer binding ($s_1 = 2$), its price fell to 0. The local formula would say $22 + 4 = 26$: wrong outside the range.

(d) Resource 2: $g = -(1,-2,2)^\top = (-1,2,-2)^\top$: $4-\delta \geq 0$, $2+2\delta \geq 0$, $3-2\delta \geq 0 \Rightarrow -1 \leq \delta \leq \frac{3}{2}$, $y_2 = 2$. Resource 3: s_3 basic, $g = e_3$: $3+\delta \geq 0 \Rightarrow \delta \geq -3$, no upper limit, $y_3 = 0$ (adding market capacity is worthless, the market limit does not bind).

Gotcha

The row of a *basic slack* also limits the range (resource 2 above is cut by the s_3 row at $\frac{3}{2}$). The sign of g: minus the displayed column. "Same basis" never means "same plan": the plan is $B^{-1}b'$. Beyond the range the slope changes, often to 0.

18 Profit change

Practice

Given. Same tableau. (a) Profit of x_1 becomes $4+\varepsilon$. (b) Profit of x_2 becomes $3+\varepsilon$. (c) In the reading example of Part V, the profit of nonbasic x_2 becomes $2+t$.

Your task. Ranges, and what changes inside them.

Solution

(a) Add $\varepsilon \times (x_1$ row): $z = 22 + 4\varepsilon + (-1-\varepsilon)s_1 + (-2+\varepsilon)s_2$. Conditions $-1-\varepsilon \leq 0$, $-2+\varepsilon \leq 0$: $-1 \leq \varepsilon \leq 2$, i.e. $c_1 \in [3,6]$. Plan $(4,2)$ unchanged, value $22+4\varepsilon$, $y = (1+\varepsilon, 2-\varepsilon, 0)$ moves.
(b) $z = 22 + 2\varepsilon + (-1+\varepsilon)s_1 + (-2-2\varepsilon)s_2$: $\varepsilon \leq 1$, $\varepsilon \geq -1$: $c_2 \in [2,4]$, value $22+2\varepsilon$.
(c) Nonbasic: $r_2 = 0 \to t$; stays optimal iff $t \leq 0$. Any increase makes x_2 enter (it was already a tie).

Gotcha

A basic profit change moves y and *every* reduced cost: recheck all of them. For a basic variable the range often has no upper end (raising its profit never makes it leave) but here both ends are finite because two slacks compete. At an endpoint, two plans are optimal.

19 New activity

Practice

Given. Running example. A new product $u \geq 0$ uses 1 h of resource 1 and 2 h of resource 2, is not limited by the market, and earns γ.

Your task. (a) For which γ is the old plan still optimal? (b) For $\gamma = 6$ find the new optimum.

Solution

(a) $a = (1, 2, 0)^\top$, $y^\top a = 1 + 4 = 5$, $r_u = \gamma - 5$. Old plan optimal iff $\gamma \leq 5$.
(b) $\gamma = 6$: $r_u = 1$. $B^{-1}a = (1-2,\ -1+4,\ 1-4)^\top = (-1, 3, -3)^\top$, column $-B^{-1}a = (1, -3, 3)^\top$: $x_1 = 4 + u$, $x_2 = 2 - 3u$, $s_3 = 3 + 3u$. Only x_2 limits: $u = \frac{2}{3}$, x_2 leaves. $u = \frac{2}{3} + \frac{1}{3}s_1 - \frac{2}{3}s_2 - \frac{1}{3}x_2$; $x_1 = \frac{14}{3} - \frac{2}{3}s_1 + \frac{1}{3}s_2 - \frac{1}{3}x_2$; $s_3 = 5 - x_2$; $z = \frac{68}{3} - \frac{2}{3}s_1 - \frac{8}{3}s_2 - \frac{1}{3}x_2$. New optimum $(x_1, x_2, u) = (\frac{14}{3}, 0, \frac{2}{3})$, $z^\star = \frac{68}{3} \approx 22.67$, $y = (\frac{2}{3}, \frac{8}{3}, 0)$. Check: $2 \cdot \frac{14}{3} + \frac{2}{3} = 10$, $\frac{14}{3} + \frac{4}{3} = 6$.

Gotcha

$y^\top a$ is the cost of the resources the unit consumes, priced at shadow prices; a product is worth making iff its profit beats that. A new column needs no new row and no new slack.

20 Simultaneous changes

Practice

Given. Running example; resource 1 becomes $10 + \delta$ and the profit of x_1 becomes $4 + \varepsilon$ at the same time.

Your task. Range where the basis stays optimal, and $z^\star(\delta, \varepsilon)$. Is the total change the sum of the separate changes?

Solution

Feasibility depends on δ only: $-3 \leq \delta \leq 2$. Optimality depends on ε only: $-1 \leq \varepsilon \leq 2$. Basis optimal on the rectangle. Value $c_B(\varepsilon)^\top B^{-1} b(\delta) = (4 + \varepsilon)(4 + \delta) + 3(2 - \delta) = 22 + \delta + 4\varepsilon + \varepsilon\delta$. Not additive: the term $\varepsilon\delta$ appears because the extra profit applies to a changed quantity of x_1.

Part VIII

Theory: what to state and how to prove it

21 Proof skeletons you must be able to write

Method

- **Weak duality**: insert $y^\top Ax$; two inequalities, each using one feasibility and one sign.
- **Final row certifies optimality**: the identity $c^\top x = z_0 + \sum r_j x_j$ holds for every solution of $Ax = b$; with $x \geq 0$ and $r \leq 0$, $c^\top x \leq z_0$; the BFS attains z_0.
- **Uniqueness from strict signs**: $c^\top x = z_0$ forces every $r_j x_j = 0$, so $x_N = 0$, then $x_B = B^{-1}b$.
- **Ratio test**: $x_B(\theta) = x_B + \theta d_B$ coordinate by coordinate; only $d_{B(k)} < 0$ bounds θ; $Ad = 0$

keeps the equations; gain $c^\top d = r_j$ because $c^\top d = c_j + c_B^\top d_B = c_j - c_B^\top B^{-1} A_j$.

- **Shadow price**: b does not appear in r, so only feasibility can break; value $c_B^\top B^{-1}(b+\Delta b) = z^\star + y^\top \Delta b$.
- y **is dual optimal**: $r \le 0$ on original columns gives $A^\top y \ge c$; on slack columns gives $y \ge 0$; $b^\top y = z^\star$; weak duality.
- **Complementary slackness**: $b^\top y - c^\top x = y^\top(b - Ax) + x^\top(A^\top y - c)$, a sum of nonnegative products; zero gap iff every product is zero; feasibility plus equal values is optimality.
- **Unbounded $\Rightarrow$ dual infeasible**: a dual-feasible y would give the upper bound $b^\top y$.
- **Termination**: strict increase without degeneracy, finitely many bases.
- **Converse of optimality needs nondegeneracy**: with $r_j > 0$ and $\theta_{\max} > 0$ there is a strictly better point; at a degenerate point $\theta_{\max}$ may be 0 and the converse fails.

Practice

Given. The tableau $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$ of the running example.

Your task. (a) Prove from the tableau alone that 22 is the optimal value and that the optimum is unique. (b) Prove that $y = (1, 2, 0)$ is dual feasible using only the reduced costs. (c) State precisely what is wrong with "an optimal BFS always has all reduced costs ≤ 0".

Solution

(a) The z row holds for every solution of the equations. A feasible point has $s_1, s_2 \ge 0$, so $z = 22 - s_1 - 2s_2 \le 22$; the point $(4, 2, 0, 0, 3)$ is feasible with $z = 22$. Uniqueness: $z = 22$ forces $s_1 = 2s_2 = 0$, hence $s_1 = s_2 = 0$, and the rows then give $x_1 = 4$, $x_2 = 2$, $s_3 = 3$: one point.
(b) $r_{s_1} = -y_1 = -1 \le 0$ and $r_{s_2} = -y_2 = -2 \le 0$ give $y_1, y_2 \ge 0$; s_3 basic gives $y_3 = 0 \ge 0$. For the original columns, $r_{x_1} = r_{x_2} = 0$ (basic) means $c_j = y^\top A_j$: $2y_1 + y_2 = 4$ and $y_1 + y_2 + y_3 = 3$ hold with equality, so $A^\top y \ge c$. Dual feasible; $b^\top y = 22$ matches, so it is dual optimal.
(c) It is true for a *nondegenerate* optimal BFS. At a degenerate optimum, a basis representing the point may have $r_j > 0$ with $\theta_{\max} = 0$ (the improving direction leaves the feasible set immediately); another basis of the same point passes the test. Example: $x + s = 0$, $\max x$: basis $\{s\}$ shows $r_x = 1 > 0$ at the optimal point 0.

Part IX

Quick checks: true or false, one line each

22 The bank

Practice

Your task. Decide, with the one-line reason.

1. A BFS has exactly m nonzero coordinates.
2. Every basis gives a BFS.
3. If a feasible basis has all nonbasic reduced costs ≤ 0, its solution is optimal.
4. If a BFS is optimal, every basis of it has reduced costs ≤ 0.

5. A positive reduced cost always allows a strictly better feasible point.
6. A zero nonbasic reduced cost at a nondegenerate optimum gives infinitely many optima.
7. A tableau with all reduced costs ≤ 0 is optimal.
8. An unbounded feasible region implies no finite optimum.
9. If the primal is unbounded, the dual is infeasible.
10. If the primal is infeasible, the dual is unbounded.
11. At an optimal pair, a tight primal row has a positive dual variable.
12. At an optimal pair, a positive primal variable has a tight dual row.
13. Changing b changes the reduced costs.
14. Changing c changes the optimal plan of the current basis.
15. The shadow price is the slope of $z^\star$ in b_i for every δ.
16. Inside the range of b_i, the optimal plan is unchanged.
17. The effect of two simultaneous changes is the sum of the separate effects.
18. Removing a redundant constraint can change the optimal value.
19. A new activity changes the old reduced costs.
20. The slack columns of the final tableau show B^{-1}.

Solution

1 F, at most m (degeneracy). 2 F, $B^{-1}b$ may have negatives. 3 T, the objective row bounds every feasible point. 4 F, only if nondegenerate. 5 F, $\theta_{\max}$ may be 0 (degenerate) or the direction may be unbounded (then infinitely better, still "better"): the false part is the degenerate case. 6 T, move along the direction up to $\theta_{\max} > 0$. 7 F, the constants must also be ≥ 0. 8 F, $\max -x_1$, $x_1 \geq 0$. 9 T, a feasible y would bound the primal. 10 F, the dual may also be infeasible. 11 F, both y_i and s_i may be 0. 12 T, $x_j t_j = 0$. 13 F, r does not involve b. 14 F, $B^{-1}b$ does not involve c; only the value and the reduced costs change. 15 F, only inside the range. 16 F, the plan is $B^{-1}b'$ and moves; the basis is unchanged. 17 F, there is a bilinear term. 18 F, the feasible set is unchanged. 19 F, it adds one reduced cost. 20 F, they show $-B^{-1}$ (equation-style tableau).